

Assignment 1

Probability measure on a set of random variables

Student:

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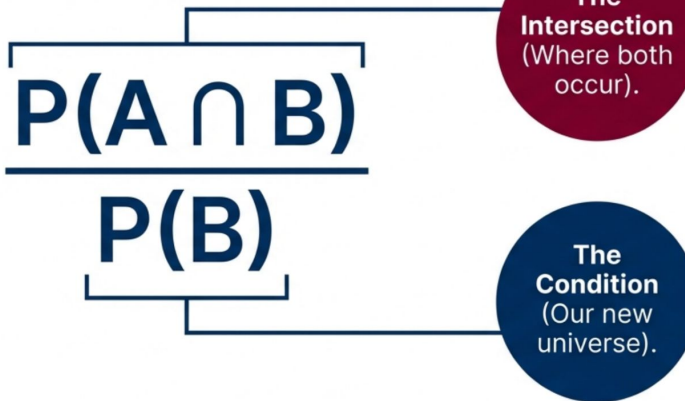
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Conditional Probability

Conditional probability is the likelihood of an event **A** occurring given that another event **B** has already happened. It is the fundamental tool for quantifying "revised uncertainty."

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$


The Intersection
(Where both occur).

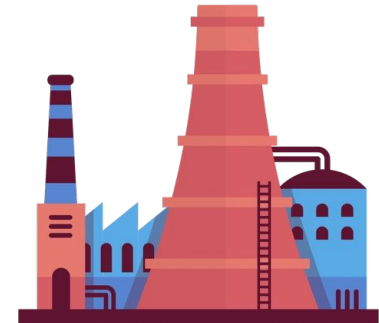
The Condition
(Our new universe).

For any two events A and B where $P(B) > 0$.

Conditional Probability

In a plant, complex components are assembled on two different assembly lines, A and A'. Line A uses older equipment than A', so it is somewhat slower and less reliable. Suppose that on a given day line A assembles 8 components, of which 2 have been identified as defective (B) and 6 as non-defective (B'), while A' produces 1 defective component and 9 non-defective ones. This information is summarized in the accompanying table.

Condition	Defective (B)	Non-Defective (B')
Line A	2	6
Line A'	1	9



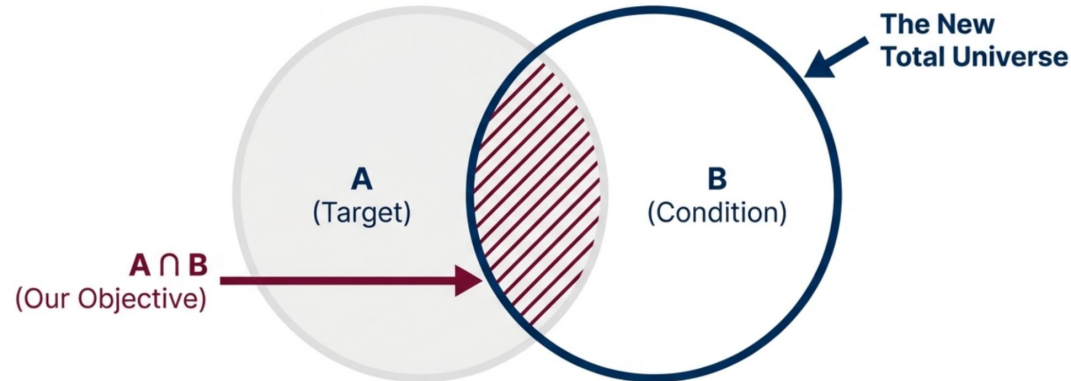
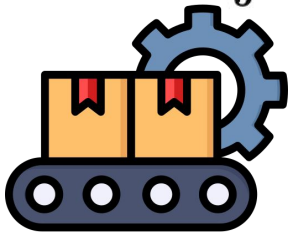
Conditional Probability

If a component is selected at random, the probability that it comes from line A is:

$$P(\text{component selected is from line A}) = P(A) = \frac{N(A)}{N} = \frac{8}{18} = 0.44$$

However, if we are told that the selected component is defective, our information changes. Now we only consider the 3 defective components. Among those defective components 2 came from line A

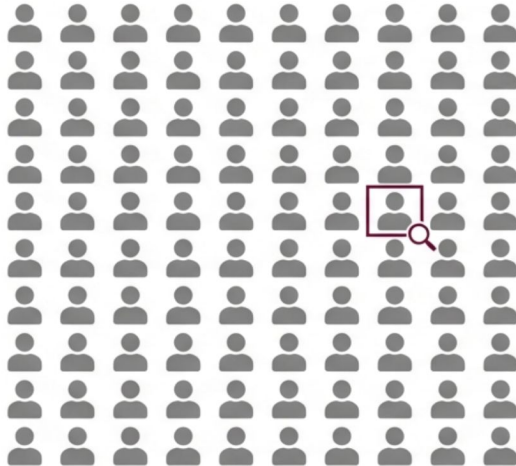
$$P(A | B) = \frac{2}{3} = \frac{2/18}{3/18} = \frac{P(A \cap B)}{P(B)}$$



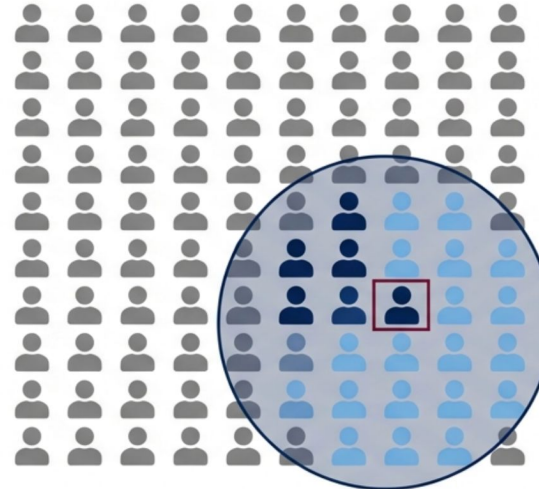
Bayes' Theorem

Describes how to update the probability of an event when new information becomes available. It connects prior beliefs with observed evidence to produce a posterior probability.

Prior: Before evidence, the probability of selecting a specific target is low.



Posterior: Evidence (The Spotlight) shrinks the universe. The target probability increases relative to this new, smaller group.



Bayes' Theorem

Let $A_1, A_2, \dots, A_k$ set of mutually exclusive and exhaustive events with prior probabilities $P(A_i)$, $i = 1, \dots, k$. Then, for any other event B such that $P(B) > 0$ the posterior probability of A_j given that B has occurred is:

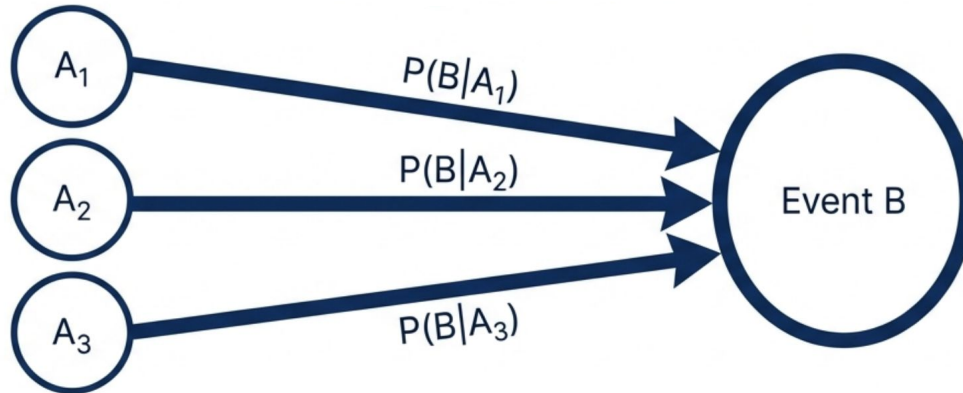
$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Law of Total Probability

Let $A_1, \dots, A_k$ be mutually exclusive and exhaustive events. Then, for any other event B

$$P(B) = P(B | A_1)P(A_1) + \dots + P(B | A_k)P(A_k)$$

$$= \sum_{i=1}^k P(B | A_i)P(A_i)$$



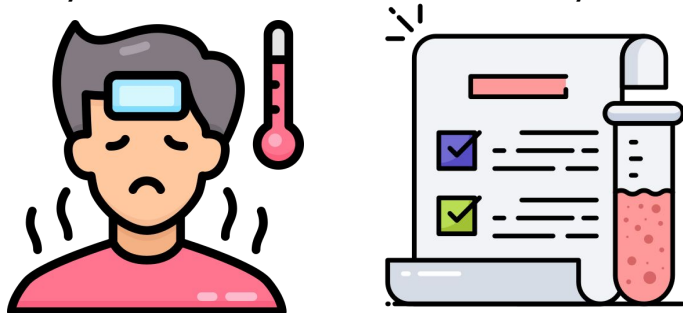
Example

A rare disease affects **1 out of every 1,000 adults**. A diagnostic test has been developed with the following properties:

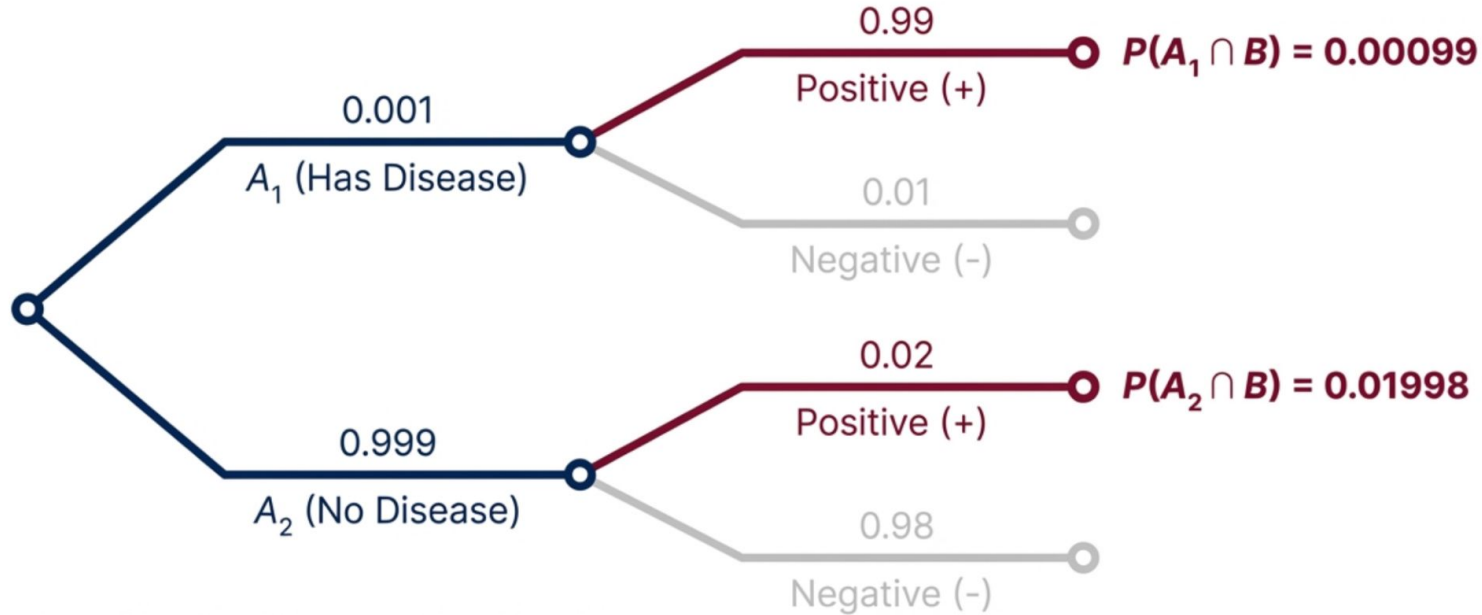
- If a person **has the disease**, the test is positive **99%** of the time.
- If a person **does not have the disease**, the test is positive **2%** of the time.

A randomly selected individual takes the test and the result is **positive**.

What is the probability that the individual actually has the disease?



Example



$$P(A_1 | B) = \frac{\text{True Positives}}{\text{Total Positives}} = \frac{0.00099}{0.02097} \quad 4.7\%$$

Gracias